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12 Most Common Types of Cyberattacks

22–28 minutes

What is a cyberattack?

A **cyberattack** is an attempt by cybercriminals, hackers or other digital adversaries to access a computer network or system, usually for the purpose of altering, stealing, destroying or exposing information.

Cyberattacks can target a wide range of victims from individual users to enterprises or even governments. When targeting businesses or other organizations, the hacker's goal is usually to access sensitive and valuable company resources, such as intellectual property (IP), customer data or payment details.

1. Malware
2. Denial-of-Service (DoS) Attacks
3. Phishing
4. Spoofing
5. Identity-Based Attacks
6. Code Injection Attacks
7. Supply Chain Attacks
8. Social Engineering Attacks
9. Insider Threats
10. DNS Tunneling
11. IoT-Based Attacks
12. AI-Powered Attacks

Expert Tip

If you're a small or medium-sized business get current stats and dive deeper into why your organization size can be at risk for cyberattacks

Read: Most common cyberattacks on SMBs

1. Malware

Malware — or malicious software — is any program or code that is created with the intent to do harm to a computer, network or server. Malware is the most common type of cyberattack, mostly because this term encompasses many subsets such as ransomware, trojans, spyware, viruses, worms, keyloggers, bots, cryptojacking, and any other type of malware attack that leverages software in a malicious way.

Type	Description
Ransomware	In a ransomware attack, an adversary encrypts a victim's data and offers to provide a decryption key in exchange for a payment. Ransomware attacks are usually launched through malicious links delivered via phishing emails, but unpatched vulnerabilities and policy misconfigurations are used as well.
Fileless Malware	Fileless malware is a type of malicious activity that uses native, legitimate tools built into a system to execute a cyber attack. Unlike traditional malware, fileless malware does not require an attacker to install any code on a target's system, making it hard to detect.
Spyware	Spyware is a type of unwanted, malicious software that infects a computer or other device and collects information about a user's web activity without their knowledge or consent.
Adware	Adware is a type of spyware that watches a user's online activity in order to determine which ads to show them. While adware is not inherently malicious, it has an impact on the performance of a user's device and degrades the user experience.
Trojan	A trojan is malware that appears to be legitimate software disguised as native operating system programs or harmless files like free downloads. Trojans are installed through social engineering techniques such as phishing or

Type	Description
	bait websites. The zeus trojan malware, a variant, has the goal accessing financial information and adding machines to a botnet.
Worms	A worm is a self-contained program that replicates itself and spreads its copies to other computers. A worm may infect its target through a software vulnerability or it may be delivered via phishing or smishing. Embedded worms can modify and delete files, inject more malicious software, or replicate in place until the targeted system runs out of resources.
Rootkits	Rootkit malware is a collection of software designed to give malicious actors control of a computer network or application. Once activated, the malicious program sets up a backdoor exploit and may deliver additional malware. Bootkits take this a step further by infecting the master boot prior to the operating system booting up, making them harder to detect.
Mobile Malware	Mobile malware is any type of malware designed to target mobile devices. Mobile malware is delivered through malicious downloads, operating system vulnerabilities, phishing, smishing, and the use of unsecured WiFi.
Exploits	An exploit is a piece of software or data that opportunistically uses a defect in an operating system or an app to provide access to unauthorized actors. The exploit may be used to install more malware or steal data.
Scareware	Scareware tricks users into believing their computer is infected with a virus. Typically, a user will see scareware as a pop-up warning them that their system is infected. This scare tactic aims to persuade people into installing fake antivirus software to remove the “virus.” Once this fake antivirus software is downloaded, then malware may infect your computer.
Keylogger	Keyloggers are tools that record what a person types on a device. While there are legitimate and legal uses for keyloggers, many uses are malicious. In a keylogger attack, the keylogger software records every keystroke on

Type	Description
	the victim's device and sends it to the attacker.
Botnet	Botnet is a network of computers infected with malware that are controlled by a bot herder. The bot herder is the person who operates the botnet infrastructure and uses the compromised computers to launch attacks designed to crash a target's network, inject malware, harvest credentials, or execute CPU-intensive tasks.
MALSPAM	Malicious malware (MALSPAM) delivers malware as the malicious payload via emails containing malicious content, such as viruses or malware-infected attachments.
Wiper Attack	A wiper attack is designed to permanently delete or corrupt data on targeted systems. They are often observed in geopolitical conflicts and in a hacktivism context.

2. Denial-of-service (DoS) attacks

A Denial-of-Service (DoS) attack is a malicious, targeted attack that floods a network with false requests in order to disrupt business operations.

In a DoS attack, users are unable to perform routine and necessary tasks, such as accessing email, websites, online accounts or other resources that are operated by a compromised computer or network. While most DoS attacks do not result in lost data and are typically resolved without paying a ransom, they cost the organization time, money and other resources in order to restore critical business operations.

The difference between DoS and Distributed Denial of Service (DDoS) attacks has to do with the origin of the attack. DoS attacks originate from just one system while DDoS attacks are launched from multiple systems. DDoS attacks are faster and harder to block than DOS attacks because multiple systems must be identified and neutralized to halt the attack.

3. Phishing

Phishing is a type of cyberattack that uses email, SMS, phone, social media, and social engineering techniques to entice a victim to share sensitive information — such as passwords or account numbers — or to download a malicious file that will install viruses on their computer or phone.

Common phishing attacks include:

Type	Description
Spear Phishing	Spear-phishing is a type of phishing attack that targets specific individuals or organizations typically through malicious emails. The goal of spear phishing is to steal sensitive information such as login credentials or infect the targets' device with malware.
Whaling	A whaling attack is a type of social engineering attack specifically targeting senior or C-level executive employees with the purpose of stealing money or information, or gaining access to the person's computer in order to execute further cyberattacks.
SMiShing	Smishing is the act of sending fraudulent text messages designed to trick individuals into sharing sensitive data such as passwords, usernames and credit card numbers. A smishing attack may involve cybercriminals pretending to be your bank or a shipping service you use.
Vishing	Vishing, a voice phishing attack, is the fraudulent use of phone calls and voice messages pretending to be from a reputable organization to convince individuals to reveal private information such as bank details and passwords.

4. Spoofing

Spoofing is a technique through which a cybercriminal disguises themselves as a known or trusted source. In so doing, the adversary is able to engage with the target and access their systems or devices with the ultimate goal of stealing information, extorting money or installing malware or other harmful software on the device.

Spoofing can take different forms, which include:

Type	Description
Domain Spoofing	Domain spoofing is a form of phishing where an attacker impersonates a known business or person with fake website or email domain to fool people into the trusting them. Typically, the domain appears to be legitimate at first glance, but a closer look will reveal subtle differences.

Type	Description
Email Spoofing	Email spoofing is a type of cyberattack that targets businesses by using emails with forged sender addresses. Because the recipient trusts the alleged sender, they are more likely to open the email and interact with its contents, such as a malicious link or attachment.
ARP Spoofing	Address Resolution Protocol (ARP) spoofing or ARP poisoning is a form of spoofing attack that hackers use to intercept data. A hacker commits an ARP spoofing attack by tricking one device into sending messages to the hacker instead of the intended recipient. This way, the hacker gains access to your device's communications, including sensitive data.

5. Identity-based attacks

Identity-driven attacks are extremely hard to detect. When a valid user's credentials have been compromised and an adversary is masquerading as that user, it is often very difficult to differentiate between the user's typical behavior and that of the hacker using traditional security measures and tools.

Some on the most common identity-based attacks include:

Type	Description
Kerberoasting	Kerberoasting is a post-exploitation attack technique that attempts to crack the password of a service account within the Active Directory (AD) environment. In a Kerberoasting attack, an adversary masquerades as an account user with a service principal name (SPN) and requests a ticket, which contains an encrypted password.
Man-in-the-Middle (MITM) Attack	A man-in-the-middle attack is a type of cyberattack in which an attacker eavesdrops on a conversation between two targets with the goal of collecting personal data, passwords or banking details, and/or to convince the victim to take an action such as changing login credentials, completing a transaction or initiating a transfer of funds.

Type	Description
Pass-the-Hash Attack	Pass the hash (PtH) is a type of attack in which an adversary steals a “hashed” user credential and uses it to create a new user session on the same network. It does not require the attacker to know or crack the password to gain access to the system. Rather, it uses a stored version of the password to initiate a new session.
Golden Ticket Attack	In a golden ticket attack, adversaries attempt to gain unlimited access to an organization’s domain by accessing user data stored in Microsoft Active Directory. The attacker exploits vulnerabilities in the Kerberos identity authentication protocol, allowing them to bypass authentication methods.
Silver Ticket Attack	A silver ticket is a forged authentication ticket often created when an attacker steals an account password. A forged service ticket is encrypted and enables access to resources for the specific service targeted by the silver ticket attack.
Credential Harvesting	In credential harvesting, cybercriminals gather user credentials — such as user IDs, email addresses, passwords, and other login information — en masse to then access systems, gather sensitive data, or sell it in the dark web.
Credential Stuffing	Credential stuffing attacks work on the premise that people often use the same user ID and password across multiple accounts. Therefore, possessing the credentials for one account may be able to grant access to other, unrelated account.
Password Spraying	The basics of a password spraying attack involve a threat actor using a single common password against multiple accounts on the same application. This avoids the account lockouts that typically occur when an attacker uses a brute force attack on a single account by trying many passwords.
Brute Force Attacks	A brute force attack is uses a trial-and-error approach to systematically guess login info, credentials, and

Type	Description
	encryption keys. The attacker submits combinations of usernames and passwords until they finally guess correctly.
Downgrade Attacks	Downgrade attacks are a cyberattack where adversaries take advantage of a system's backward compatibility to force it into less secure modes of operation, such as forcing a user to go into a HTTP version of a website instead of HTTPS.

6. Code injection attacks

Code injection attacks consist of an attacker injecting malicious code into a vulnerable computer or network to change its course of action. There are multiple types of code injection attacks:

Type	Description
SQL Injection	A SQL Injection attack leverages system vulnerabilities to inject malicious SQL statements into a data-driven application, which then allows the hacker to extract information from a database. Hackers use SQL Injection techniques to alter, steal or erase application's database data.
Cross-Site Scripting (XSS)	Cross Site Scripting (XSS) is a code injection attack in which an adversary inserts malicious code within a legitimate website. The code then launches as an infected script in the user's web browser, enabling the attacker to steal sensitive information or impersonate the user. Web forums, message boards, blogs and other websites that allow users to post their own content are the most susceptible to XSS attacks.
Malvertising	Malvertising attacks leverage many other techniques, such as SEO poisoning, to carry out the attack. Typically, the attacker begins by breaching a third-party server, which allows the cybercriminal to inject malicious code within a display ad or some element thereof, such as banner ad copy, creative imagery or video content. Once clicked by a website visitor, the corrupted code within the ad will install

Type	Description
	malware or adware on the user's computer.
Data Poisoning	Data poisoning is a type of cyberattack in which an adversary intentionally compromises a training dataset used by an Artificial Intelligence or Machine Learning model to manipulate the operation of that model. When dataset is manipulated during the training phase, the adversary can introduce biases, intentionally create erroneous outputs, introduce vulnerabilities, or otherwise influence predictive capabilities of the model.

7. Supply chain attacks

A supply chain attack is a type of cyberattack that targets a trusted third-party vendor who offers services or software vital to the supply chain. **Software supply chain attacks** inject malicious code into an application in order to infect all users of an app, while **hardware supply chain attacks** compromise physical components for the same purpose. Software supply chains are particularly vulnerable because modern software is not written from scratch: rather, it involves many off-the-shelf components, such as third-party APIs, open source code and proprietary code from software vendors.

Social engineering is a technique where attackers use psychological tactics to manipulate people into taking a desired action. Through the use of powerful motivators like love, money, fear, and status, attackers can gather sensitive information that they can later use to either extort the organization or leverage such information for a competitive advantage.

Examples of social engineering attacks include:

Attack	Description
Pretexting	In pretexting, attackers get access to information, a system, or a user by posing a false scenario that gains the victim's trust. This includes posing as an investment banker, an HR employee, or even an IT specialist.

Attack	Description
Business email compromise (BEC)	In a BEC attack, attackers assume the identity of a trusted user to trick employees or company customers into making payments or sharing data, among other things.
Disinformation campaign	Disinformation campaigns are deliberate efforts to spread false information, especially for political or war-related reasons. Adversaries leverage social media networks that reach vast audiences to amplify false narratives through the prolific use of bots and fake accounts, creating a false sense of consensus.
Quid Pro Quo	Using the quid pro quo technique, attackers target users with an offer to pay in exchange for a product or service.
Honeytrap	Honeytrap attacks target individuals looking for love or friendship on dating apps/websites. Attackers create fake profiles and leverage the relationship built overtime to trick the victim into giving them money, information, or access to their network to install malware.
Tailgating/ Piggybacking	Tailgating, also known as piggybacking, is a type of attack conducted in person by tagging along behind a company employee and asking them to hold the door open. Once the adversary is inside the facilities, they physically try to steal or destroy sensitive information.

9. Insider threats

IT teams that solely focus on finding adversaries external to the organization only see half the picture. Insider threats are internal actors such as current or former employees that pose danger to an organization because they have direct access to the company network, sensitive data, and IP as well as knowledge of business processes, company policies, or other information that would help carry out such an attack.

Internal actors that pose a threat to an organization tend to be malicious in nature. Some motivators include financial gain in exchange for selling confidential information on the dark web and/or emotional coercion such as

the ones used in social engineering tactics. But some insider threats are not malicious in nature — instead, they are negligent. To combat this, organizations should implement a comprehensive cybersecurity training program that teaches stakeholders to be aware of any potential attacks, including those potentially performed by an insider.

10. DNS tunneling

DNS Tunneling is a type of cyberattack that leverages domain name system (DNS) queries and responses to bypass traditional security measures and transmit data and code within the network.

Once infected, the hacker can freely engage in command-and-control activities. This tunnel gives the hacker a route to unleash malware and/or to extract data, IP or other sensitive information by encoding it bit by bit in a series of DNS responses.

DNS tunneling attacks have increased in recent years, in part because they are relatively simple to deploy. Tunneling toolkits and guides are even readily accessible online through mainstream sites like YouTube.

11. IoT-based attacks

An internet of things (IoT) attack is any cyberattack that targets an IoT device or network. Once compromised, the hacker can assume control of the device, steal data, or join a group of infected devices to create a botnet to launch DoS or DDoS attacks

Given that the number of connected devices is expected to grow rapidly, cybersecurity experts expect IoT infections to grow as well. Furthermore, the deployment of 5G networks, which will fuel the use of connected devices, may also lead to an uptick in attacks.

Expert Tip

IoT devices include traditional endpoints — such as computers, laptops, mobile phones, tablets, and servers — as well as nontraditional devices, such as printers, cameras, appliances, smart watches, health trackers, navigation systems, smart locks, or smart thermostats.

12. AI-powered attacks

As AI and ML technology improves, the number of use cases has also

increased. Just as cybersecurity professionals leverage AI and ML to protect their online environments, attackers also leverage these tools to get access to a network or steal sensitive information.

Examples of AI-powered cyberattacks include:

Attack	Description
Adversarial AI/ML	Adversarial artificial intelligence and machine learning seek to disrupt the operations of AI and ML systems by manipulating or misleading them. They can do this by introducing inaccuracies in training data.
Dark AI	Dark AI is specifically engineered to leverage the benefits of incorporating AI and ML technology to exploit vulnerabilities. Dark AI usually goes unnoticed until the damage is done.
Deepfake	Deepfakes are AI-generated forgeries that appear very real and have the potential to reshape public opinion, damage reputations, and even sway political landscapes. These can come in the form of fake images, videos, audio, or more.
AI-generated social engineering	Attackers create fake chatbots or virtual assistants capable of having human-like interactions and engaging in conversations with users to get them to provide sensitive information.

How to protect against cyberattacks

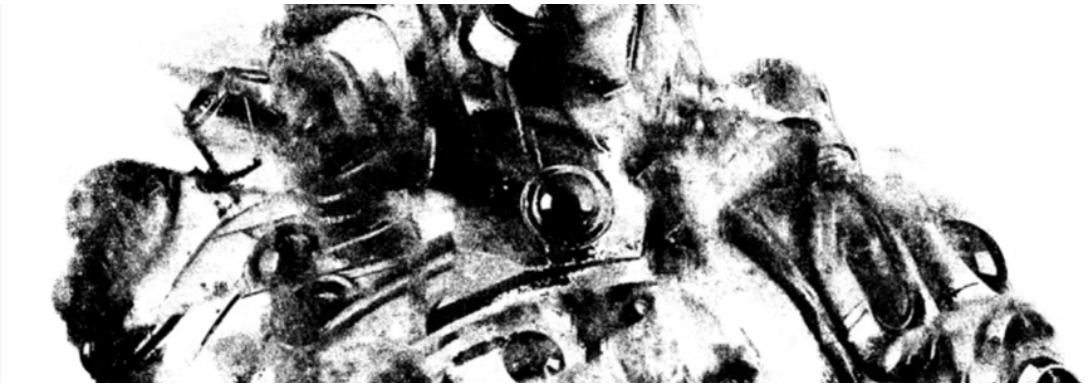
A comprehensive cybersecurity strategy is absolutely essential in today's connected world. From a business perspective, securing the organization's digital assets has the obvious benefit of reducing the risk of loss, theft, destruction, and the potential need to pay a ransom to regain control of company data or systems. Employing a comprehensive cybersecurity strategy can help organizations prevent or quickly remediate cyberattacks and minimize the impact of these events on business operations.

Finally, when an organization takes steps to deter adversaries, it protects the brand from the reputational harm often associated with cyberattacks — especially those that involve the loss of customer data.

Below are some recommendations we offered in the CrowdStrike 2024

Global Threat Report to help organizations improve their security posture and ensure cybersecurity readiness:

- **Protect All Workloads**: You must secure all critical areas of enterprise risk, including endpoints and cloud workloads, identity, and data.
- **Know Your Adversary**: CrowdStrike Falcon® Adversary Intelligence identifies today's bad actors and exposes their playbooks to enable security teams to proactively optimize preventions, strengthen defenses, and accelerate incident response.
- **Be Ready When Every Second Counts**: Security teams of all sizes must invest in speed and agility for their daily and tactical decision-making by automating preventive detection, investigation, and response workflows with integrated cyber threat intelligence directly observed from the front lines.
- **Adopt Zero Trust**: Because today's global economy requires data to be accessible from anywhere at any time, it is critical to adopt a Zero Trust model. CrowdStrike® Identity Protection connects your machines to identity and data to deliver full Zero Trust protection.
- **Monitor the Criminal Underground**: Adversaries congregate to collaborate using a variety of hidden messaging platforms and dark web forums. Leverage digital risk monitoring tools like Falcon Adversary Intelligence to monitor imminent threats to your brand, identities, or data.
- **Invest in Elite Threat Hunting**: The combination of technology with expert threat hunters is absolutely mandatory to see and stop the most sophisticated threats. Top-quality managed services such as CrowdStrike Falcon® Complete and CrowdStrike Falcon® Adversary OverWatch can help you close the growing cyber skills gap with the expertise, resources, and coverage needed to augment your team.
- **Build a Comprehensive Cybersecurity Training Program**: User awareness programs should be initiated to combat the continued threat of phishing and related social engineering techniques.



2024 CrowdStrike Global Threat Report

The **2024 Global Threat Report** unveils an alarming rise in covert activity and a cyber threat landscape dominated by stealth. Data theft, cloud breaches, and malware-free attacks are on the rise. Read about how adversaries continue to adapt despite advancements in detection technology.

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